

SYSTEMS ENGINEERING

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



Outline

- 1 Basic Concepts
- 2 Human Activities •
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



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- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



What is Systems Engineering?

- **Systems Engineering** is a discipline that studies the design, implementation, and maintenance of complex systems.
- This discipline is based on interdisciplinary fields, such as control engineering, industrial engineering, software engineering, mechanical engineering, electrical engineering, organizational studies, project management, and others.
- **Systems Engineering** is a holistic approach to engineering that focuses on how to design and manage complex systems over their lifecycle.

design



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What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
- This **discipline** is based on interdisciplinary fields, such as **control engineering**, **industrial engineering**, **software engineering**, **mechanical engineering**, **electrical engineering**, **organizational studies**, **project management**, and **others**.
- **Systems Engineering** is a **holistic** approach to **engineering** that focuses on how to **design** and manage **complex systems** over their **lifecycle**.



Systems Engineer as a Professional

- A **Systems Engineer** is a professional who is responsible for designing, implementing, and maintaining complex systems.
- A **Systems Engineer** must have a broad understanding of engineering, mathematics, science, and technology.
- A **Systems Engineer** must be able to analyze and solve problems in a systematic and holistic way.

architect
tech lead
pre-sales

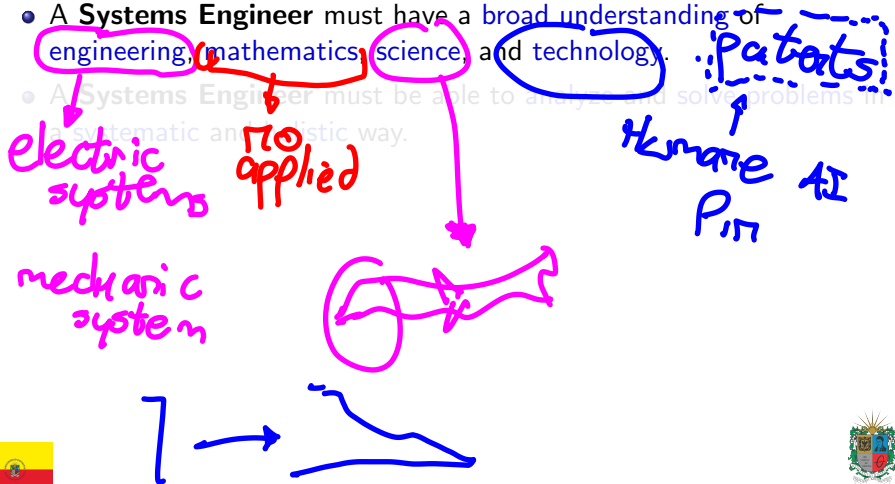
software
engineering

Academic & researchers



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Programming
Contest

Saturday
12 - 18

10 Problems

English

1 kb

100

Google

Apostol



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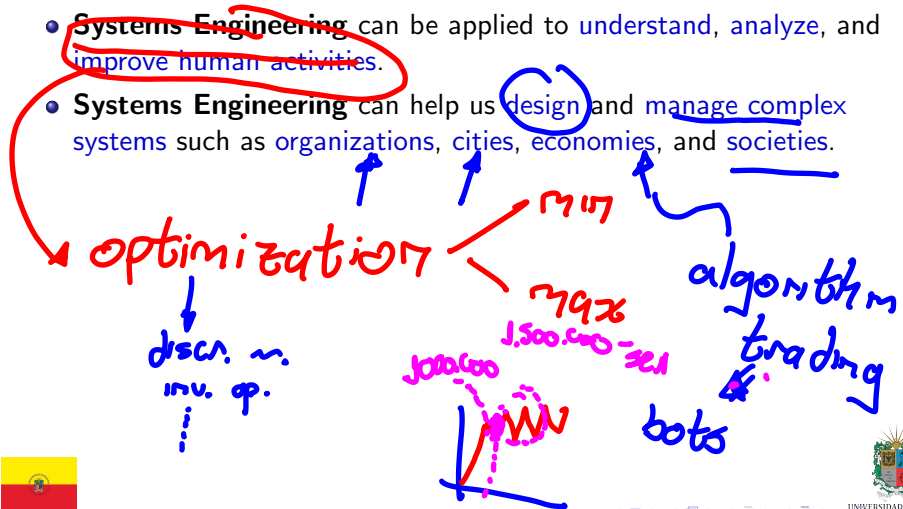


Human Activities

- **Human activities** are **complex systems** that involve **multiple components, interactions, and feedback loops**.
- **Systems Engineering** can be applied to understand, analyze, and improve human activities.
- **Systems Engineering** can help us design and manage complex systems such as organizations, cities, economies, and societies.



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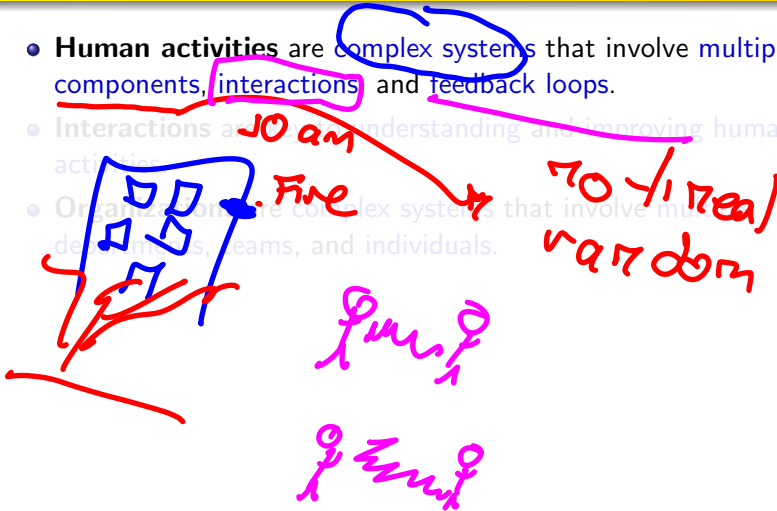


Human Activities: Complexity

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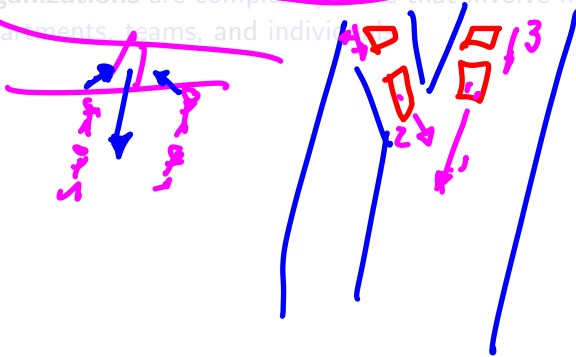
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- Organizations are complex systems that involve multiple departments, teams, and individuals.



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different points of view
 different dynamics
 not all days are the same



Organizations as Systems

- **Organizations** can be **viewed** as: a rational system, a natural system, or an open system.

- A **rational system** is a formal organization that is designed to achieve specific goals as a machine.
environment
- A **natural system** is an informal organization that is emergent and adaptive based on human interactions.
logic
- An **open system** is an organization that is interconnected with its environment and adapts to changes.
human



Organizations as Systems

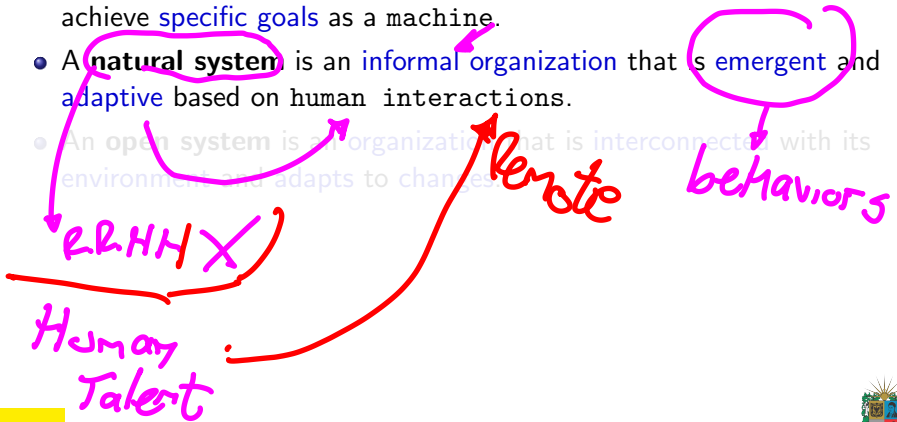
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mission
vision
goals



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clients
suppliers
competence
↳ rivals

change
plan

startup



Synergy

- **Synergy** is a simple but powerful concept: the whole is greater than the sum of its parts.
- It means the **interactions** could boost the capabilities of the parts of the **system**. Also, it lets both understand **emergent behaviors** and define improvements in **systems**.

6-degree separation

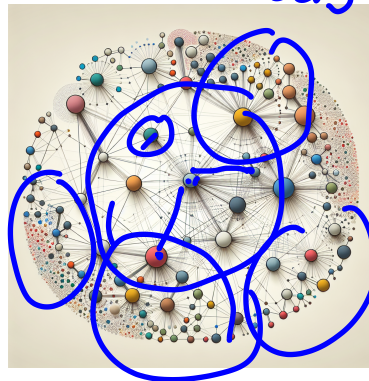


Figure: Prompt: Define a draw of clusters in social networks.



Synergy

- **Synergy** is a simple but powerful concept: **the whole is greater than the sum of its parts**.
- It means the **interactions** could **boost the capabilities** of the parts of the **system**. Also, it lets both understand **emergent behaviors** and define **improvements** in **systems**.
- One of the main concepts is the **theory of computation**. Based on **graphs**, you could define a **computational machine**.

*Algo
Turing*

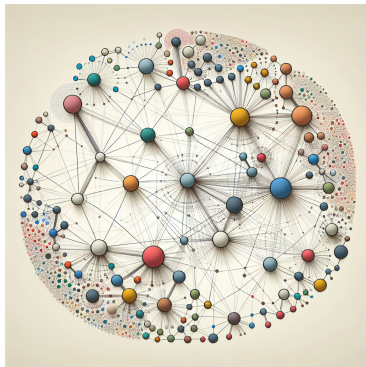


Figure: Prompt: Define a draw of clusters in social networks.



Money Ball

→ Begegnung

economist

Gravitet

20/10/11

stats

\$100,000

22 mit
str. ke

the signal and the noise

why so many predictions fail—but some don't

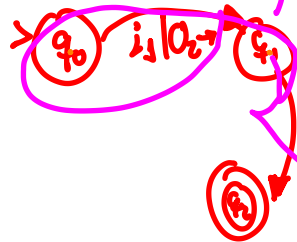
~~nate silver~~



Talking with Machines!



Algorithm

↓
DAGfinite
state
machine

- **Alan Turing** proposed a hundred years ago an **Universal Machine**, capable of take any algorithm defined as a state machine, and process it in a binary language. → **Digital**

Forty years after, Noam Chomsky proposed the formal languages based on generative grammars. Here is where the high-level programming languages appear.

0101011010

algo

0101010
input

parameter



Talking with Machines!

$\Sigma = \{a, b\}$

$G = \begin{cases} S \rightarrow aV_1 \\ V_1 \rightarrow ba|ab|V_2 \\ V_2 \rightarrow \vdots \end{cases}$

all words of
a language

tree

graph
(FSM)

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→ binary → UTM



Programming Languages

- **Programming Languages** with more capabilities, easier comprehension had been created. Also, more people start to code into specific domain programming languages.

Natural Language

Java: hungry? lunch: home;
Python:



lunch if hungry else home

Demo time!

General-purpose

CD (Raggs)

SQL

Terraform
(IaC)



Programming Languages

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Demo time!

- **Andrej Karpathy**, a leading figure at **tesla** and now at **OpenAI**, said: *Nowadays, English is the most important programming language.*

NLP

LLM code

agent intelligent

vibing code

A sand B GPS C reported D fish



Applications of Systems Engineering

- **Systems Engineering** can be applied to understand, analyze, and improve complex systems.
- **Systems Engineering** can be applied to design and manage complex systems such as organizations, cities, economies, and societies.
- **Systems Engineering** can be applied to design and manage complex systems such as software, hardware, and networks.

Handwritten notes and arrows:

- Red arrows point from the words "understand", "analyze", "improve", "design", "manage", "organizations", "cities", "economies", "societies", "software", "hardware", and "networks" in the list above to the corresponding terms in the handwritten notes below.
- A large blue oval encloses the handwritten text "IoT Robotics" and "cloud".
- Red text "IoT Robotics" is written below the blue oval.
- Red text "cloud" is written to the right of the blue oval.
- A red arrow points from the word "improve" in the first bullet point to a red "\$" symbol.
- A red arrow points from the word "design" in the third bullet point to the word "Ubiquitous" in the handwritten text below.

Handwritten text:

- Ubiquitous → IoT
- ↳ Woatras
- > 93 Countries
- > 7 Petabytes/week



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Cibernetics

~1448

- **Cibernetics** is the study of systems, control, and communication in animals, machines, and organizations.

- Cibernetics is a transdisciplinary field that combines engineering, mathematics, biology, psychology, and philosophy.

- Cibernetics is the foundation of systems engineering and information technology.

environment
auto-regulation

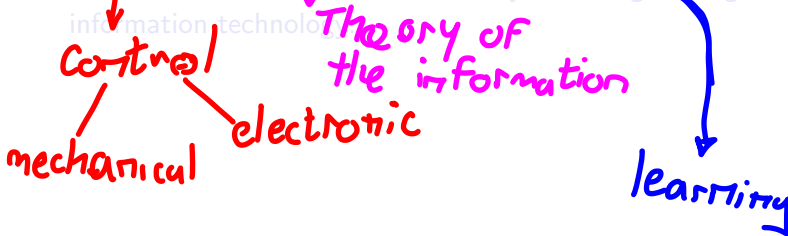


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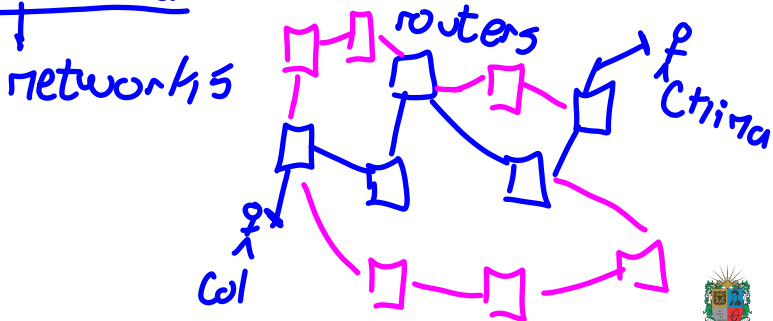
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Technology

- **Technology** is the application of scientific knowledge to solve problems and improve systems.

Temporal is the key to dealing with complex systems
such as organizations, cities, economies, and societies.

~ 2000-3000 years ago



100
WWI



Technology

- **Technology** is the *application* of scientific knowledge to solve problems and improve systems.
- **Technology** is the key to designing and managing complex systems such as organizations, cities, economies, and societies.

Computers as
tools

Comp.
science

math
phys.
bido.
Chem.



Technology: **AI**

- **Artificial Intelligence** is a *field* of computer science that studies how to design and implement intelligent agents.

Artificial Intelligence is the
 solve problems
 like a human ~

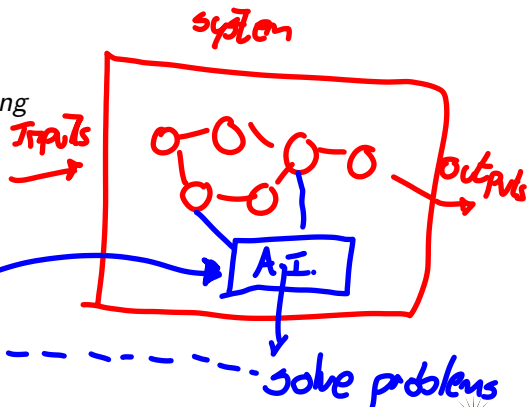
software
 program
 who solve problems
 → reasoning
 → intuition/reflex

→ maths
 → statistical
 → probabilistics
 → discrete
 → non-linear
 → biology
 → arm
 → eye-lens
 → psychology
 → learning

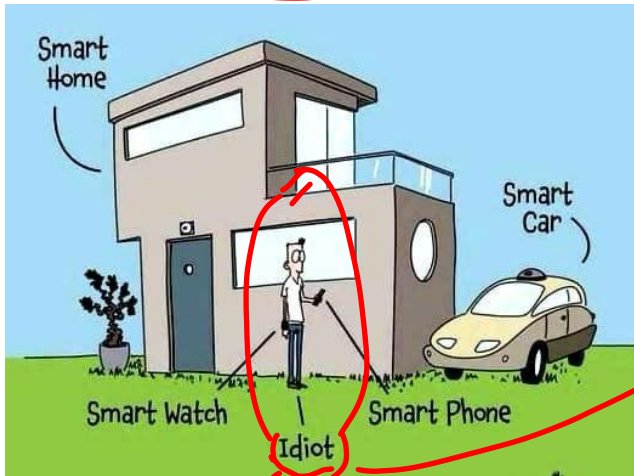


Technology: AI

- **Artificial Intelligence** is a *field* of computer science that *studies* how to *design* and *implement* intelligent agents.
- **Artificial Intelligence** is the key to *designing* and *managing* *complex systems* such as smart-organizations, smart-cities, smart-economies, and smart-societies.



Artificial Intelligence **not** as a System



Chat
GPT
↓
less
reasoning
capacity

Wall-e



Cibernetics & Technology in Systems Context

- **Cibernetics** and **technology** are the foundation of systems engineering and information technology.
- **Cibernetics** and **technology** are the foundation of artificial intelligence and smart systems.

//
sciences
control

rage



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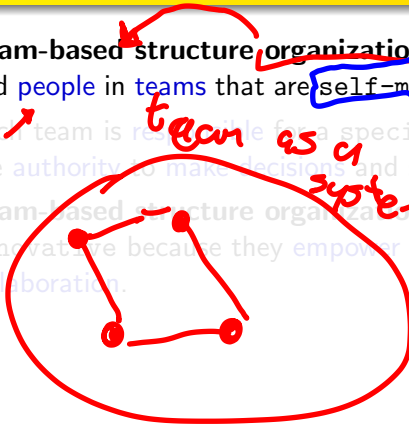


Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are **self-managed** and **cross-functional**

- Each team is responsible for a specific task or project and has the authority to make decisions and solve problems.

- Team-based structure organizations are flexible, agile, and innovative because they empower employees and encourage collaboration.

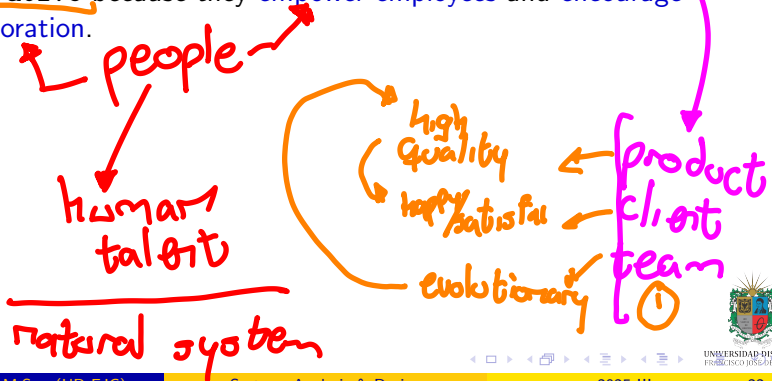


flexible
agile



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Teams as a System

- **Teams** are a **system** where collaboration (**synergy**) and communication are **key** to **success**.
- Defining and **automating** the **processes** and **procedures** within the **team** is a **challenge** to **improve team performance**.
- **Teams** are like **pieces** in a **puzzle**, where each **piece** has a **specific role** and **responsibility**.

part relations

any Ops

capabilities
skills

daily



Soft Skills

1st step

- **Soft skills** are personal attributes that enable someone to interact effectively and harmoniously with other people.

- Typical Soft Skills:

- ✓ • Communication skills (verbal and written).
- ✓ • Teamwork and collaboration.
- ✓ • Problem-solving and critical thinking.
- ✓ • Adaptability and flexibility.
- ✓ • Time management and organization.
- ✓ • Leadership and management.
- ✓ • Emotional intelligence.
- ✓ • Creativity and innovation.
- ✓ • Conflict resolution.
- ✓ • Networking and relationship building.
- ✓ • Customer service and client management.

Programming
Contest

Test of
Wartegg

note



Computer Analyst

system

● Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

statistical

↓
descriptive

● Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
- Providing support and training to end users.



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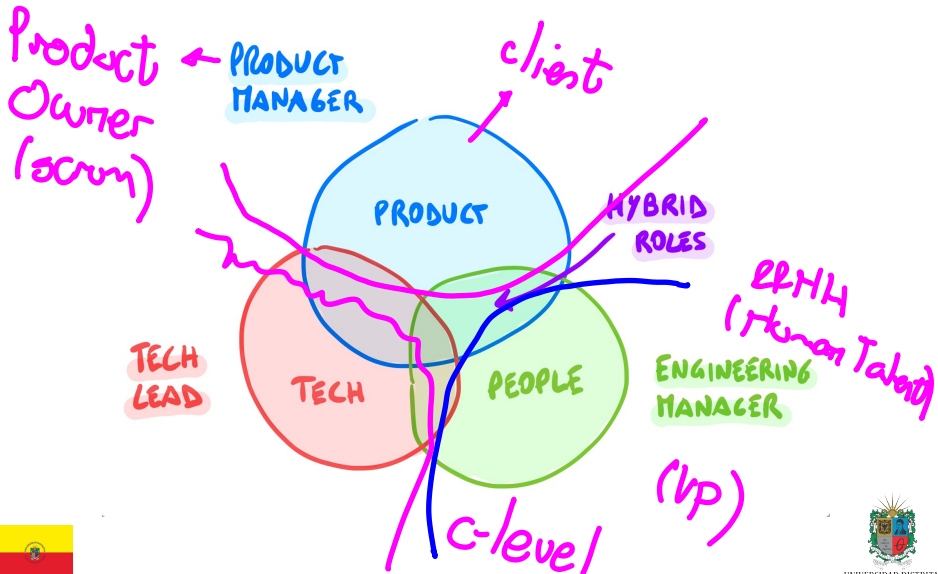
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startups

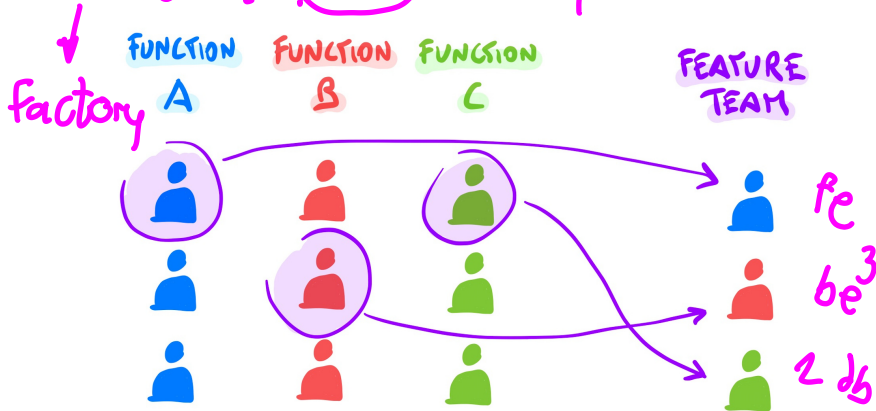


Tech Company Typical Structure



Feature Teams

Globalant → pool → study → no project



What is to be a leader? I

- **Leading** a team is **not a role**. It is a decision, you could be a **leader** anytime and anywhere.
 - ↑
 - ↑
 - Junior X
- **Teamwork culture** is pretty important. It creates habits, open communication, safety spaces for inclusion.
- **Psychological safety** is a key point to have an effective team. You could develop *technical skills*, but it is not enough.
 - Senior ✓
 - lead
- **Hierarchy** is very important. Anarchism tends to fail. Hierarchy exists by **status** and **power**.
- In a hierarchy experts lead to make better decisions. However, anyone must be careful to not leave people behind.



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daily
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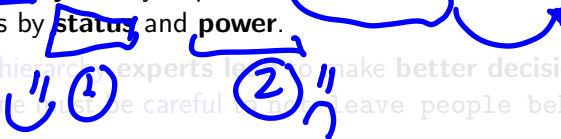
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What is to be a leader? II

- With **crystal-clear communications** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A good leader should focus on outcomes rather than outputs. This helps bring business value rather than just complete tasks.
- Failure is always an option. Learn how to deal with setbacks; do not punish — just fix and learn.
- Some believe you're born a leader, while others think that a leader can be developed over time. Either way, context and the desire for self-growth are vital.
- Making ethical decisions is key; it leads to taking the right and better actions.



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The Outliers



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What is to be a leader? III

- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A good leader builds trusting relationships and has the emotional intelligence to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: Curiosity, Courage, and Commitment.



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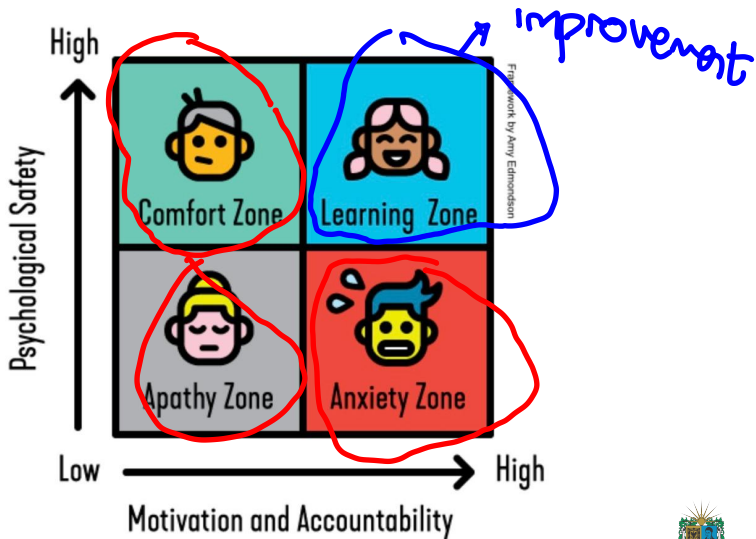


What is to be a leader? III

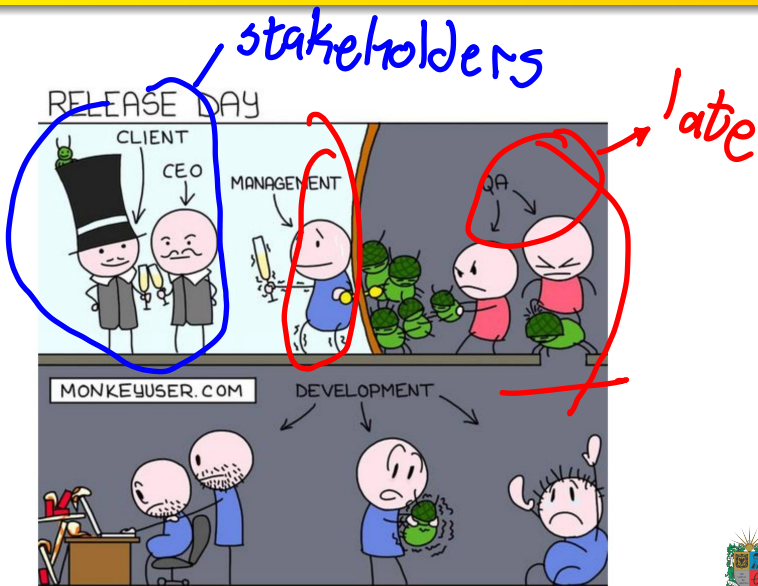
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Working Zones



Real World!



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

